

From Support Fit to Exact Shared Anatomical Volume Mapping

Technical Progress Report Aligned with `representation.pdf`

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Current result. Fifteen accepted normal-shoe instances now have validated, fold-free tetrahedral volumes connected to one canonical anatomical volume by exact forward and inverse piecewise-affine maps. All 15 passed independent B3 geometry validation and Checkpoint 11-C round-trip validation. The semantic anatomical fibers needed to express the exact coordinate as (u, v, r) remain Checkpoint 11-D.

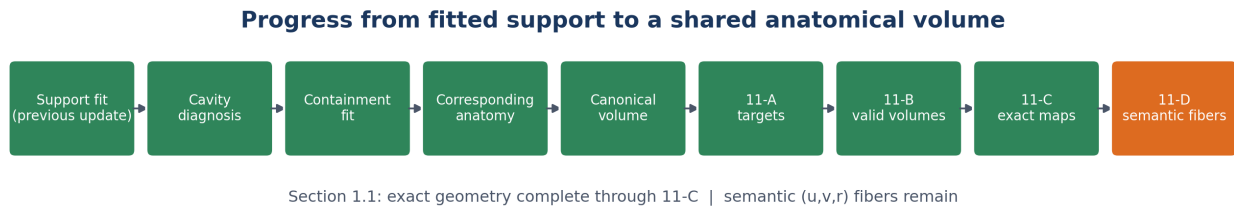


Figure 1: Implemented pipeline since the previously reported support-fit stage. Green stages are complete; orange is the next stage.

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1 Purpose, scope, and relation to the representation document

The design document `notes/representation.pdf` proposes an *anatomy-grounded volumetric footwear representation*. Its central idea is to separate a shared anatomical coordinate system from footwear geometry, whose surface topology may vary arbitrarily. For footwear instance i , it defines a fitted foot

$$F_i = F(\beta_i, \theta_i) \subset \mathbb{R}^3, \quad (1)$$

a canonical anatomical domain A around a reference foot F_0 , and a volumetric map

$$\chi_i : A \rightarrow N(F_i), \quad \Phi_i = \chi_i^{-1}. \quad (2)$$

The document writes the shared coordinate as

$$q = \Phi_i(x) = (u, v, r), \quad (3)$$

where (u, v) identifies a functional anatomical surface origin and r describes outward progress through the surrounding volume. It also asks for an approximately injective, bounded-distortion map instead of a nearest-surface heuristic:

$$0 < c \|x_1 - x_2\|_2 \leq \|\Phi_i(x_1) - \Phi_i(x_2)\|_2 \leq C \|x_1 - x_2\|_2, \quad C < \infty. \quad (4)$$

The implementation reported here realizes this proposal with a canonical tetrahedral cage, fixed tetrahedron connectivity, a finite-element harmonic r field, bounded deformation, and exact barycentric lookup. The exact internal coordinate is currently

$$q_{\text{exact}} = (k, \lambda_0, \lambda_1, \lambda_2, \lambda_3), \quad \sum_{a=0}^3 \lambda_a = 1, \quad (5)$$

where k is a canonical tetrahedron ID. This exact coordinate is implemented and validated. The remaining 11-D task is to associate it with a semantic surface chart (u, v) and an anatomical fiber parameterized by r .

1.1 Status at a glance

Implementation stage	Purpose	Document	relation	Status
Support fit	Seat an articulated SUPR foot on the detected footbed	Prerequisite Sec. 1.1	to	Previously reported
Cavity analysis	Measure forbidden collisions and local upper/side clearance	Validates F_i		Complete
Containment fit	Search anatomical shape, pitch, and placement while preserving support	Constructs fitted F_i		Complete
Anatomical surface and leg	Transfer stable anatomical identity and extend through the lower leg	Sec. 1.1 boundary construction		Complete
Canonical volume	Build A , its fixed envelope, tetrahedra, and harmonic r	Sec. 1.1 canonical domain		Complete
11-A	Transfer the computational inner-boundary target to every fitted anatomy	Desired instance boundary		Complete
11-B	Produce valid instance volumes with fixed topology and no folds	Constructs geometric χ_i		Complete
11-C	Implement exact forward/inverse queries and invalid-point classification	Evaluates χ_i, Φ_i		Complete
11-D	Associate volume points with semantic (u, v, r) fibers	Completes semantic Sec. 1.1 coordinate		Next
Sec. 1.2	Learn coverage and material extents in shared coordinates	Structured material field		Not started

Table 1: Mapping from implementation checkpoints to `representation.pdf`.

The initial geometry pipeline produced artifacts for 16 normal-profile shoes. The final shared-volume scope contains 15 shoes: `sneaker_vibe` was deliberately excluded before B3 because its source shoe geometry has an inconsistent orientation relative to the accepted normalized set. It was not repaired, loaded, or counted as a B3/11-C failure.

2 Starting point: the accepted support fit

The last reported stage placed a posed SUPR foot on a detected normalized shoe support. It established plantar seating, heel-to-toe alignment, coordinate transforms, and a reproducible SUPR shape/pose state. That result answered “where should the foot sit?”, but not “does the complete foot lie in usable shoe space?” A foot can be well supported and still cross a sidewall, toe box, upper, or collar. Steps 12 and 13 therefore treated the accepted support result as immutable input and added explicit cavity reasoning.

Figure 2 follows `sandal_1` through this transition. This shoe is useful as a running example because its support fit had many detected collisions, yet its final containment result and later tetrahedral mapping are valid.

3 Step 12: cavity collision and clearance analysis

3.1 Geometric objective

The detected footbed is the only shoe surface on which contact is allowed. Every other original shoe triangle forms the obstacle surface $\mathcal{S}_{\text{obs}}$. The ankle may leave through actual empty opening space,

but a real triangle intersection with the collar remains forbidden. This preserves open sandals and shoe entrances without adding an artificial lid.

For a foot point p , unsigned obstacle clearance is the closest-surface distance

$$d(p, \mathcal{S}_{\text{obs}}) = \min_{y \in \mathcal{S}_{\text{obs}}} \|p - y\|_2. \quad (6)$$

Because the shoe may be open and non-watertight, a global mesh sign is not assumed. Instead, local signed upper and side clearances are measured by ray intersections with the corresponding local boundary. When both measurements exist, the conservative combined clearance is

$$c(p) = \min\{c_{\text{upper}}(p), c_{\text{side}}(p)\}. \quad (7)$$

The same measurement is made at foot vertices and triangle centroids. For face f , with available sample set S_f , the conservative face value is

$$c_f = \min_{s \in S_f} c_s. \quad (8)$$

A face is signed outside when $c_f < -\varepsilon$. Exact triangle–triangle tests are also performed between the foot and obstacle surfaces. If $\mathcal{F}_{\text{coll}}$ contains the unique foot faces in exact intersections, the collision area fraction is

$$C_{\text{coll}} = \frac{\sum_{f \in \mathcal{F}_{\text{coll}}} A_f}{\sum_{f \in \mathcal{F}_{\text{foot}}} A_f}. \quad (9)$$

Area is used rather than the raw number of intersecting pairs so that a densely triangulated shoe does not appear artificially worse. Signed protrusion depth and its area-weighted energy are

$$p_s = \max(0, -c_s), \quad (10)$$

$$e_f = \frac{1}{|S_f|} \sum_{s \in S_f} p_s^2, \quad (11)$$

$$E_{\text{protrusion}} = \frac{\sum_f A_f e_f}{\sum_f A_f}. \quad (12)$$

Finally, the affected region is the union of exact collision faces and signed-outside faces:

$$\mathcal{F}_{\text{affected}} = \mathcal{F}_{\text{coll}} \cup \mathcal{F}_{\text{outside}}. \quad (13)$$

Among the final 15-shoe scope, one support fit was already clear and 14 had detected collisions. A problem status here is a diagnosis, not an execution failure.

4 Step 13: collision-aware containment fitting

4.1 What was optimized

The containment search adjusted the actual SUPR anatomical model rather than shrinking one template uniformly. Its control vector was

$$\vartheta = [\beta_0, \dots, \beta_9, \phi_{\text{ankle}}, \phi_{\text{midfoot}}, t_x, t_z]^T. \quad (14)$$

The ten β coefficients change correlated anatomical shape; the two pitch controls change ankle and midfoot articulation; and t_x, t_z alter heel-to-toe and lateral placement. Foot size was not a separate free scale. Under the anchored support scale, length, width, and height changed only through the SUPR shape parameters.

The search enforced fixed, shoe-independent constraints:

$$18 \text{ mm} \leq a(\vartheta) \leq 22 \text{ mm}, \quad a_{\text{target}} = 20 \text{ mm}, \quad (15)$$

$$|\Delta\phi_{\text{ankle}}| \leq 4^\circ, \quad |\Delta\phi_{\text{midfoot}}| \leq 4^\circ, \quad (16)$$

$$-3 \leq \beta_j \leq 3. \quad (17)$$

Every candidate had to return to a valid first contact with the stored support and retain heel, forefoot, toe, and overall support coverage. It also could not regress beyond the baseline collision-area tier or baseline signed-outside-area tier.

4.2 Implemented search mathematics

This stage did *not* minimize one arbitrary weighted sum of support, collision, and shape terms. It used a deterministic broad search, local clearance correction, and lexicographic selection.

The broad search included individual and coupled beta directions and 256 deterministic Latin-hypercube samples over $\beta_1, \dots, \beta_9$. For each template, β_0 was solved toward 18, 20, and 22 mm toe allowance. Support-valid candidates in the target band received exact collision tests, and ten beta-diverse candidates were refined.

At a local candidate, near-conflict samples form

$$\mathcal{A} = \{j \mid c_j \leq h\}, \quad (18)$$

where h is support-grid spacing. The desired clearance correction is

$$\Delta c_j^* = \max(0, -c_j + \varepsilon). \quad (19)$$

The clearance Jacobian is measured by finite differences of the clearance actually used in Eq. (7):

$$J_{jk} \approx \frac{c_j(\vartheta + \delta_k e_k) - c_j(\vartheta)}{\delta_k}. \quad (20)$$

An additional row steers toe position toward the 20 mm target. The joint update is the minimum-norm least-squares correction

$$\Delta\vartheta = J^+ \Delta c^*, \quad (21)$$

followed by bounds, exact reevaluation, local polling, and line factors $\{2, 1, \frac{1}{2}, \frac{1}{4}\}$. The final exact candidates are ordered by the following strict priorities:

$$\begin{aligned} \text{rank}(\vartheta) = & (\text{toe-band validity, toe-band distance, complete containment,} \\ & \text{affected-area tier, collision-area tier, } C_{\text{coll}}, \text{outside-area tier,} \\ & \text{maximum protrusion, } E_{\text{protrusion}}, |a - 20 \text{ mm}|, \|\beta\|_2, \\ & E_{\text{placement}}, E_{\text{support}}). \end{aligned} \quad (22)$$

Earlier entries dominate later entries; Eq. (22) is not a scalar sum. The placement-deviation diagnostic is

$$E_{\text{placement}} = \left(\frac{t_x}{h}\right)^2 + \left(\frac{t_z}{h}\right)^2 + \left(\frac{\Delta\phi_{\text{ankle}}}{4^\circ}\right)^2 + \left(\frac{\Delta\phi_{\text{midfoot}}}{4^\circ}\right)^2. \quad (23)$$

For **sandal_1**, the support-fit baseline had 859 exact intersecting pairs and $C_{\text{coll}} = 0.186662$. The selected containment result had 20.0000066 mm toe allowance, zero exact collisions, zero signed-outside area, and status **contained_target_fit**. Across the final 15-shoe scope, five shoes achieved **contained_target_fit**; ten retained the honest status **residual_target_fit**. A residual fit is still a plausible authoritative anatomy. It means that footwear material lying inside that anatomy must later be reported outside the mathematical footwear-support domain rather than silently mapped.

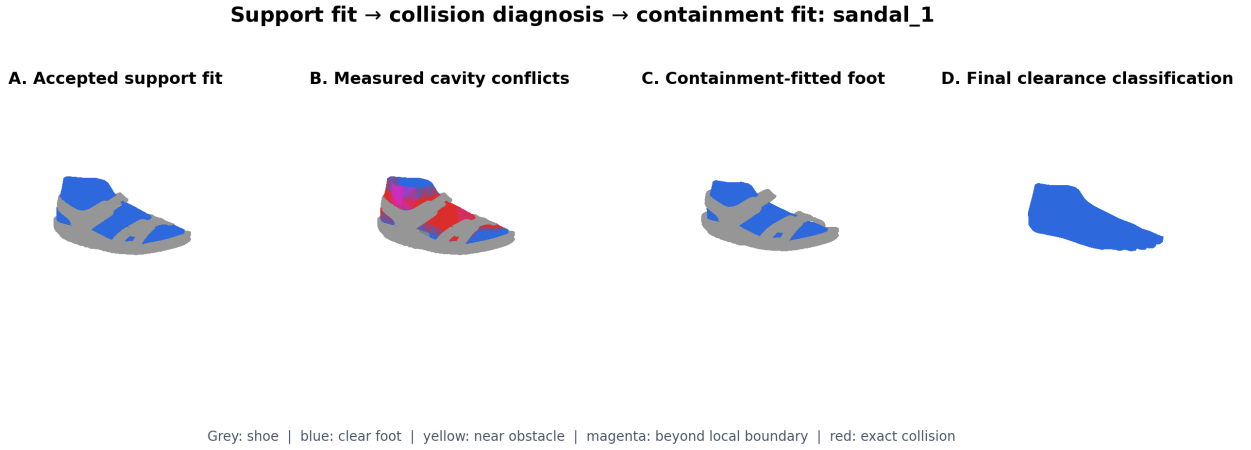


Figure 2: Actual saved geometry for **sandal_1**. The support fit was valid but collided with the sandal. The containment search found a supported, realistic-size SUPR foot with no remaining collision for this case.

5 Steps 14–16: corresponding foot-and-lower-leg anatomy

The representation requires anatomical meaning to remain stable even though physical vertex positions change. SUPR supplies stable native vertex and face IDs. A deterministic two-level subdivision then creates a dense foot containing 4,151 vertices and 8,240 faces; the first 266 vertices remain the accepted native fitted foot exactly.

It is important to separate a *mesh vertex* from a general *surface point*. The anatomical mesh consists of filled triangles, not only their corner vertices. A point may therefore lie at a triangle corner, on an edge, or anywhere inside the filled triangle. Suppose such a point p_0 lies on canonical anatomical face $f_m = (V_{0,a}, V_{0,b}, V_{0,c})$. Its position can be recorded by the face ID m and three barycentric weights $\gamma = (\gamma_a, \gamma_b, \gamma_c)$:

$$p_0 = \gamma_a V_{0,a} + \gamma_b V_{0,b} + \gamma_c V_{0,c}, \quad \gamma_a + \gamma_b + \gamma_c = 1. \quad (24)$$

Every fitted SUPR anatomy uses the same anatomical vertex IDs and face IDs. Thus face m in shoe i has the same anatomical identity but moved corners $(V_{i,a}, V_{i,b}, V_{i,c})$. The corresponding surface point is obtained without a new nearest-point search:

$$p_i = \gamma_a V_{i,a} + \gamma_b V_{i,b} + \gamma_c V_{i,c}. \quad (25)$$

The face ID and weights stay fixed; only the three corner positions change. Therefore correspondence means “the same anatomical address”, not “the same three-dimensional position”. This is the basic mechanism later used by Checkpoint 11-A.

The lower-leg stage keeps the dense foot immutable, attaches a near-neutral full-body SUPR shank at the fitted ankle, and searches limited ankle pitch/roll and tightly bounded shape correction. Empty entrance space remains allowed; exact contact with actual collar triangles is measured and retained when natural limits are reached. The joined extended surface contains 6,951 vertices and 13,832 faces with stable IDs for the foot, ankle transition, and lower leg. Of the final 15 shoes, 11 had a clear lower-leg exit and four retained residual collar intersections.

Anatomical construction with stable correspondence

A. Dense foot surface regions



B. Lower-leg exit and collar



C. Joined foot and lower leg



Figure 3: Actual anatomical outputs for `sandal_1`. Region colors remain tied to the same anatomy while the lower leg and ankle transition extend the domain for boots and shoe collars.

6 Step 17: canonical tetrahedral anatomical volume

6.1 Authoritative anatomy versus computational boundary

The 6,951-vertex fitted surface is authoritative and is never repaired in place. The canonical neutral foot contains native toe self-overlap, which is unsuitable as a closed tetrahedral boundary. A separate computational copy is therefore repaired, subdivided, reattached to the unchanged dense lower leg, and closed at the knee. Two-way surface face-and-barycentric maps preserve its connection to the authoritative anatomy.

These are two different meshes with two different jobs:

- The **authoritative anatomical surface** preserves the original anatomical meaning and SUPR correspondence. Its extended foot-and-lower-leg version has 6,951 vertices and 13,832 faces.
- The **computational boundary** is the clean, closed surface used to build the tetrahedral volume. It has been repaired near problematic toes and includes a computational knee cap. It is not a replacement for the authoritative anatomy.

Because the meshes have different vertices and faces, a computational vertex is not assumed to equal an anatomical vertex. Instead, each computational vertex is attached to a precisely recorded point on the authoritative surface. This attachment is explained step by step in Section 7.

This produces a computational inner boundary with 8,224 vertices and 16,444 faces. A fixed outer envelope has 642 vertices and 1,280 faces. Gmsh fills the shell with 14,440 volume vertices and 59,081 positively oriented first-order tetrahedra while preserving the supplied boundary vertices and triangles.

The outer surface is a fourth-order superellipsoid:

$$\sum_{d \in \{x,y,z\}} \left| \frac{x_d - c_d}{a_d} \right|^4 = 1, \quad c = (0.45, -0.825, 0), \quad a = (0.85, 1.225, 0.48). \quad (26)$$

The same envelope is kept bitwise fixed for every shoe. Consequently the shoe geometry does not redefine the coordinate system.

6.2 Finite-element harmonic outward coordinate

The canonical scalar r minimizes Dirichlet energy

$$E_r(r) = \frac{1}{2} \int_A \|\nabla r\|_2^2 dV, \quad (27)$$

equivalently satisfying

$$\nabla^2 r = 0 \quad \text{in } A, \quad r = 0 \text{ on anatomical skin}, \quad r = 1 \text{ on the outer envelope}. \quad (28)$$

The computational knee cap is a closure, not anatomy; its non-rim vertices receive the natural boundary condition rather than fixed $r = 0$. For linear tetrahedral basis functions N_a , the local stiffness matrix is

$$K_{ab}^{(k)} = V_k \nabla N_a \cdot \nabla N_b, \quad (29)$$

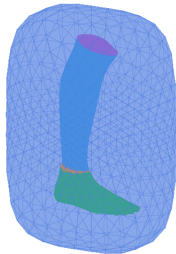
and after assembling the global matrix, the free values solve

$$K_{II} r_I = -K_{IB} r_B. \quad (30)$$

This r is a normalized harmonic layer coordinate, not physical distance in millimetres. It is stored canonically and later transported with exact tetrahedral coordinates.

Canonical tetrahedral anatomical volume

A. Closed inner boundary and fixed envelope



B. Canonical sagittal section colored by harmonic r

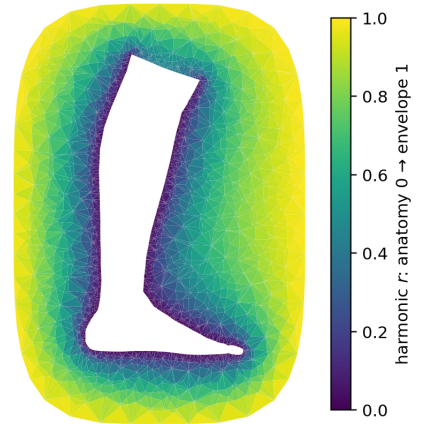


Figure 4: Canonical domain A . Left: labeled inner boundary inside the translucent fixed envelope. Right: an actual sagittal tetrahedral section colored by the solved harmonic r field.

7 Checkpoint 11-A: fitted computational-boundary targets

7.1 Notation used in the construction

The following symbols distinguish the anatomical mesh, the computational mesh, and a shoe-specific target. The two subscripts have simple meanings: i identifies the shoe, whereas j identifies a computational-boundary vertex.

Symbol	Meaning
i	Shoe or fitted-foot instance index, for example <code>sandal_1</code> .
j	Vertex index on the computational inner boundary, $j = 1, \dots, 8224$.
a, b, c	Three authoritative anatomical vertex indices forming one anatomical face.
m	ID of an authoritative anatomical triangular face.
$V_{0,a}$	Position of anatomical vertex a on the neutral canonical reference anatomy.
$V_{i,a}$	Position of that same anatomical vertex a after fitting the anatomy to shoe i .
C_j	Position of computational-boundary vertex j in the canonical reference.
P_j	Closest point on the canonical authoritative anatomical surface to C_j ; it is not a newly inserted vertex and is not random.
γ_j	Three barycentric weights recording where P_j lies in its anatomical face.
$T_{i,j}$	Desired position of computational vertex j on the fitted anatomy for shoe i ; this is the 11-A target vertex.

Table 2: Variables used to construct a Checkpoint 11-A target.

7.2 Step 1: attach the canonical computational mesh to the canonical anatomy

Consider one canonical computational vertex C_j . It lies near the canonical authoritative anatomy, but it generally does not coincide with one of its 6,951 vertices because the two meshes have different topology and resolution. We search the complete filled triangular anatomical surface for the closest point:

$$P_j = \arg \min_{p \in S_0^{\text{anat}}} \|C_j - p\|_2, \quad (31)$$

where S_0^{anat} is the canonical authoritative anatomical surface. “A point on the surface” means any location inside any filled triangle, including its edges and corners; it does not mean only an existing mesh vertex.

Suppose the resulting point P_j lies on anatomical face $f_m = (V_{0,a}, V_{0,b}, V_{0,c})$. We calculate weights satisfying

$$P_j = \gamma_{j,a} V_{0,a} + \gamma_{j,b} V_{0,b} + \gamma_{j,c} V_{0,c}, \quad \gamma_{j,a} + \gamma_{j,b} + \gamma_{j,c} = 1. \quad (32)$$

For vertex C_j , we save the face ID m and these three weights. Together they form an exact anatomical address for P_j . Multiple computational vertices may attach to different points in the same anatomical triangle; they then share the face ID but have different weights.

7.3 Step 2: evaluate the same anatomical address on fitted shoe i

The fitted foot is the desired anatomy for shoe i . It has the same authoritative vertex IDs and face IDs as the canonical anatomy, although its vertices have moved due to shape, pose, and placement. Therefore, the canonical face $(V_{0,a}, V_{0,b}, V_{0,c})$ corresponds directly to fitted face $(V_{i,a}, V_{i,b}, V_{i,c})$.

Checkpoint 11-A reuses the stored face ID and the same weights on that fitted face:

$$T_{i,j} = \gamma_{j,a} V_{i,a} + \gamma_{j,b} V_{i,b} + \gamma_{j,c} V_{i,c}. \quad (33)$$

Thus $T_{i,j}$ means: “where computational vertex j should be placed to represent the same anatomical location on fitted shoe i .” No closest-point search is repeated on the fitted foot. Repeating that search could allow the attachment to jump to a nearby but anatomically different toe. Reusing the face ID and weights preserves identity.

This operation is repeated for all 8,224 computational vertices:

$$\underbrace{C_j}_{\text{canonical computational vertex}} \longrightarrow \underbrace{(m, \gamma_j)}_{\text{saved anatomical address}} \longrightarrow \underbrace{T_{i,j}}_{\text{target on fitted anatomy}}. \quad (34)$$

The collection $T_i = \{T_{i,1}, \dots, T_{i,8224}\}$ is a new target mesh. Its vertex positions follow fitted anatomy i , while its faces, vertex numbering, and connectivity remain those of the canonical computational boundary. The actual fitted anatomical surface is neither replaced nor edited.

7.4 What 11-A has and has not accomplished

At this point the desired inner boundary is known, so tetrahedral movement can begin in Checkpoint 11-B. However, 11-A has moved only the target surface positions. It has not yet moved the surrounding tetrahedral interior, fixed the outer envelope, or guaranteed that the target surface is geometrically valid.

A clean reference proxy is not guaranteed to remain intersection-free after Eq. (33). Moving a fixed triangle network onto a narrower or differently articulated toe configuration can make non-neighboring triangles cross. Therefore 11-A is a *corresponding desired boundary*, not a cleaned final mesh. Within the final 15-shoe scope, six targets were directly **ready**; nine were **ready_requires_untangling**. Only foot-skin intersections were allowed at this stage; bridge, lower-leg, or knee-cap intersections would fail validation.

Checkpoint 11-A: corresponding target, not yet a final valid volume

A. Target over authoritative anatomy

B. Foot-region detail

C. Computational target only



Blue: computational target | grey: immutable fitted anatomy | purple: knee cap | red: target self-intersection

Figure 5: Checkpoint 11-A for **sandal_1**. The overlay validates anatomical proximity and correspondence. Red triangles mark target self-intersections that the later B3 computational-surface repair must resolve.

8 Checkpoint 11-B: valid shoe-specific volume deformation

Checkpoint 11-A tells us where each of the 8,224 inner-boundary vertices *wants* to go, but it does not move the tetrahedra around the foot. Checkpoint 11-B performs that movement.

One can picture the canonical volume as flexible foam between two surfaces. Its inner surface is the computational foot, its outer surface is the fixed envelope, and its interior is divided into 59,081 numbered tetrahedra. For shoe i , 11-B moves the inner surface toward the target T_i , keeps the outer surface fixed, and moves the intervening vertices smoothly. The tetrahedron numbers and connections never change. Therefore, tetrahedron k retains the same canonical identity even though its physical corner positions change for each shoe.

The movement cannot be performed by simply replacing every C_j with $T_{i,j}$. Doing so would move only the skin and could flatten, invert, or overlap the tetrahedra attached to it. The boundary and interior must be moved together while checking that the volume remains a valid one-to-one domain.

8.1 Notation used in the deformation

Symbol	Meaning
Q	All canonical volume-vertex positions before deformation. On the inner boundary, these positions include the computational vertices C_j .
T_i	Complete 11-A inner-boundary target for shoe i ; its j th position is $T_{i,j}$.
U	Displacement of volume vertices from their canonical positions.
X	Current or final positions of all volume vertices during deformation.
I and B	Interior/free volume vertices and prescribed boundary vertices, respectively.
K	Finite-element stiffness matrix that couples neighboring volume vertices and produces a smooth displacement.
α	B2 progress from the canonical volume (0) toward the complete smooth target deformation (1).
β	B3 progress from the canonical inner surface (0) toward the complete 11-A target (1).
k	Canonical tetrahedron ID.
F_k	Local deformation matrix of tetrahedron k from canonical to current shape.
$J_k = \det(F_k)$	Local signed volume ratio. A positive value preserves orientation; zero is flat; a negative value is inside-out.
$\sigma_{\min}, \sigma_{\max}$	Smallest and largest local stretch factors of F_k .
κ	Condition number $\sigma_{\max}/\sigma_{\min}$, measuring anisotropic distortion.
h	One fixed characteristic spacing of the fitted surface, used to normalize lengths.

Table 3: Main variables used by Checkpoint 11-B.

The required result has a fixed outer envelope, positive cell orientation, bounded distortion, no inner self-intersection, no inner-outer intersection, and only a bounded deviation from the authoritative fitted anatomy.

8.2 B1: strict input validation

B1 cross-validates the canonical volume, 11-A target, and authoritative fitted anatomy: schema and stage names, shoe identity, profile, array dimensions, finite values, topology and geometry digests, correspondence arrays, surface fidelity, recorded intersection pairs, and frozen-envelope containment. In simple terms, B1 confirms that the canonical foam, desired inner surface, and fitted anatomy belong together and have not been silently changed. It is a diagnostic gate and performs no deformation.

8.3 B2: smooth finite-element continuation

For inner-boundary vertex j , the required displacement is the arrow from its canonical position C_j to its 11-A target $T_{i,j}$. Collecting those arrows for the complete boundary gives

$$U_{\partial,\text{inner}} = T_i - Q_{\partial,\text{inner}}, \quad U_{\partial,\text{outer}} = 0. \quad (35)$$

Here $Q_{\partial,\text{inner}}$ is the collection of canonical computational-boundary positions, so its j th entry is C_j . The second equation says that every outer-envelope vertex has exactly zero displacement.

The unknown displacements of the interior vertices are then chosen to vary smoothly between the moving inner surface and the stationary outer surface. The same tetrahedral stiffness matrix used for harmonic r gives

$$K_{II}U_I = -K_{IB}U_B. \quad (36)$$

U_B contains the already prescribed inner and outer boundary movements, and U_I contains the unknown interior movements. K_{II} describes how free interior vertices are connected to one another, while K_{IB} describes how they are connected to the boundary. Solving the equation makes the interior follow the boundary smoothly.

Because this baseline is linear, continuation evaluates

$$X(\alpha) = Q + \alpha U, \quad 0 \leq \alpha \leq 1. \quad (37)$$

At $\alpha = 0$, $X(0) = Q$, so nothing has moved. At $\alpha = 1$, B2 has attempted the complete smooth movement. Intermediate values move every vertex by the same fraction of its computed displacement.

Trials begin with a step of 0.25. A failed trial is discarded and the step is halved until the 10^{-4} minimum. For tetrahedron k , B2 monitors the orientation ratio

$$J_k(\alpha) = \frac{V_k(X(\alpha))}{V_k(Q)}. \quad (38)$$

The ratio $J_k = 1$ means that cell k retains its canonical signed volume, a small positive value means that it has become nearly flat, and a non-positive value means that it is flat or inside-out. B2 rejects non-finite vertices, $J_k < 10^{-6}$, degenerate inner faces, envelope escape, inner self-intersections, and inner-outer intersections. If a trial fails, B2 returns to the preceding valid state and tries a smaller progress step. Its last valid state is only a safe starting point, not necessarily the final answer. For `sandal_1`, B2 stopped safely at $\alpha = 0.769043$ rather than flattening or inverting the limiting cells.

8.4 B3: localized bounded-distortion optimization

B2 may stop because an exact 11-A target would make a few toe triangles cross or a few attached tetrahedra become too flat. B3 resolves this by allowing small corrections to the *computational copy only*. The authoritative fitted anatomy and saved 11-A target are never edited.

B3 starts with the local surface region containing the 11-A intersection faces and weak boundary tetrahedra. Vertices outside that region follow the current target exactly. The initial region is expanded by two neighboring face rings and may grow deterministically; a full-surface fallback exists if a local region is insufficient. The optimizer then advances the desired boundary toward $\beta = 1$, while keeping each permitted correction within $0.5h$. Once the corrected surface reaches complete target progress, it is fixed and the tetrahedral interior is solved between that surface and the unchanged outer envelope.

The target continuation is

$$T_i(\beta) = (1 - \beta)Q_{\partial, \text{inner}} + \beta T_i, \quad 0 \leq \beta \leq 1. \quad (39)$$

Thus $T_i(0)$ is the canonical inner boundary and $T_i(1) = T_i$ is the complete 11-A target. The parameter β lets B3 solve a sequence of smaller, safer problems rather than demanding the full difficult shape in one jump.

For a canonical tetrahedron and its deformed instance,

$$D_k^0 = [Q_1 - Q_0 \quad Q_2 - Q_0 \quad Q_3 - Q_0], \quad (40)$$

$$D_k = [X_1 - X_0 \quad X_2 - X_0 \quad X_3 - X_0], \quad (41)$$

$$F_k = D_k(D_k^0)^{-1}, \quad J_k = \det(F_k). \quad (42)$$

The implemented normalized objective is

$$E = 10E_{\text{target}} + E_{\text{surface}} + E_{\text{FEM}} + 0.1E_{\text{distortion}} + E_{\text{barriers}}. \quad (43)$$

For active inner vertices $\mathcal{I}$, global surface area weights w_j , all computational-boundary edges $\mathcal{E}$, and surface resolution h ,

$$E_{\text{target}} = \sum_{j \in \mathcal{I}} w_j \left\| \frac{X_j - T_{i,j}(\beta)}{h} \right\|_2^2, \quad (44)$$

$$E_{\text{surface}} = \frac{1}{|\mathcal{E}|} \sum_{(a,b) \in \mathcal{E}_{\text{incident}}} \left\| \frac{(X_b - X_a) - (T_{i,b}(\beta) - T_{i,a}(\beta))}{h} \right\|_2^2, \quad (45)$$

$$E_{\text{FEM}} = \frac{(X - B(\beta))^T K (X - B(\beta))}{V_A}, \quad (46)$$

$$E_{\text{distortion}} = \sum_k \hat{V}_k \left[\frac{\|F_k\|_F^2}{3J_k^{2/3}} - 1 + 0.1(\log J_k)^2 \right]. \quad (47)$$

Here $B(\beta)$ is the smooth FEM baseline, V_A is canonical shell volume, and $\hat{V}_k$ are normalized canonical cell-volume weights. The distortion expression is scale-aware AMIPS with explicit volume control.

The determinant, collision-gap, envelope, and correction-cap barriers share a smooth one-sided form. For a lower-bounded value $x > f$ activated below a , let $y = (x - f)/(a - f)$. Then

$$b_{\text{lower}}(x) = \begin{cases} -(1 - y)^2 \log y, & f < x < a, \\ 0, & x \geq a, \\ +\infty, & x \leq f. \end{cases} \quad (48)$$

The determinant barrier activates below 0.1 and has emergency floor 10^{-6} . Collision protection activates below $0.25h$ and keeps a numerical separation above $10^{-4}h$. Computational-boundary correction is capped at $0.5h$. Barrier multipliers increase through 1, 10, 100, 100. Analytical gradients are supplied to deterministic CPU L-BFGS-B; exact triangle intersection tests, rather than the approximate barrier, remain the acceptance authority.

Final acceptance requires

$$\begin{aligned}
&\beta = 1, \quad \min_k J_k \geq 0.02, \quad \min_k \sigma_{\min}(F_k) \geq 0.05, \\
&\max_k \sigma_{\max}(F_k) \leq 5, \quad qqquad \max_k \kappa(F_k) \leq 20, \\
&\text{zero inner self-intersections,} \quad \text{zero inner-outer intersections,} \\
&\max_j \|X_j - T_{i,j}\| \leq 0.5h, \quad qqquad 0.5 \leq A_f(X)/A_f(T_i) \leq 2.0,
\end{aligned} \tag{49}$$

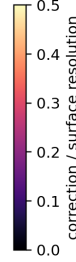
plus exact outer-envelope equality and region-specific reverse fidelity to the immutable fitted anatomy.

Checkpoint 11-B on the difficult sandal_1 case

A. B2 safe warm start ($\alpha = 0.769$)



B. B3 boundary correction



C. Final tetrahedra: $\det(F_k)$



Figure 6: B2 and B3 on `sandal_1`. B2 preserves the last safe state. B3 uses small, localized computational-boundary corrections and produces a positive, bounded-distortion final tetrahedral volume.

8.5 Final 15-shoe result

All 15 accepted shoes reached $\beta = 1$ and passed independent post-run geometric revalidation. Four are `final_exact_target`; 11 are `final_corrected_target`; none failed. Every final mesh has zero inner self-intersections, zero inner-outer intersections, and an outer envelope bitwise equal to the canonical envelope. The fresh 15-worker CPU batch completed in 919.4 seconds of batch wall time with one numerical-library thread per worker.

For the difficult `sandal_1` case, the final minimum determinant is 0.095656, maximum condition number is 9.3634, maximum correction is $0.258h$, 275 of the 8,224 inner vertices are corrected, and no interior L-BFGS iterations were necessary after the corrected boundary FEM solve.

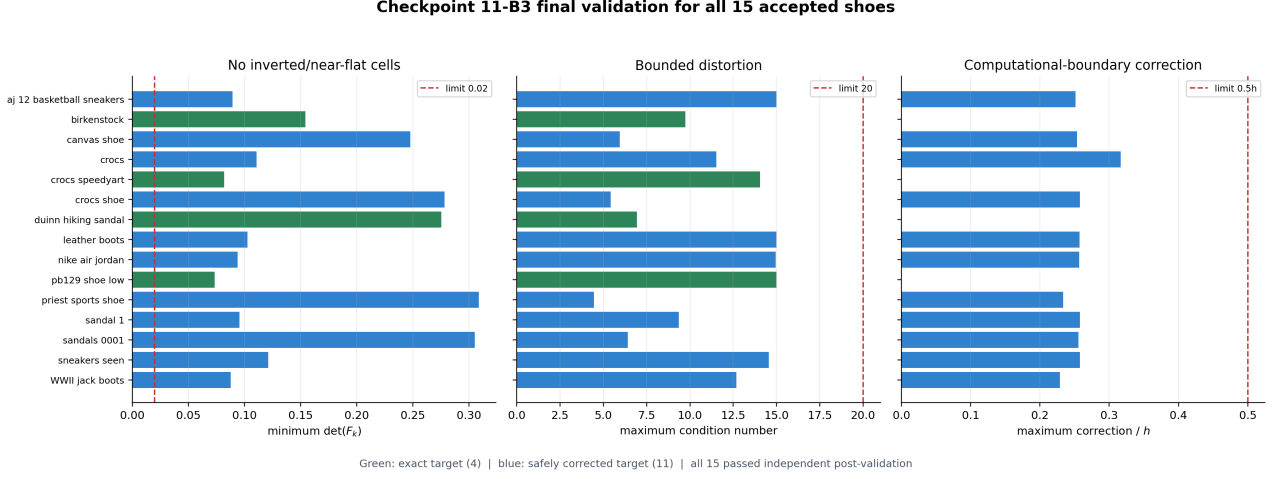


Figure 7: Per-shoe final B3 metrics. The red dashed lines are fixed acceptance limits, not fitted per-shoe thresholds. Every bar remains on the accepted side.

For one cell, the singular values of F_k give the local metric bounds

$$\sigma_{\min}(F_k)\|dq\|_2 \leq \|dx\|_2 \leq \sigma_{\max}(F_k)\|dq\|_2. \quad (50)$$

Thus the B3 singular-value and condition-number tests are the implemented local analogue of the bounded-distortion requirement in Eq. (4). Positive determinants prevent local folds; exact boundary collision checks and 11-C overlap audits address global ambiguity.

9 Checkpoint 11-C: exact forward and inverse volume maps

Checkpoint 11-C is a deterministic lookup layer over the completed B3 grids. It does not deform, repair, optimize, or remesh anything.

For canonical tetrahedron k with corners $Q_{k,0}, \dots, Q_{k,3}$, instance corners $X_{i,k,0}, \dots, X_{i,k,3}$, and barycentric weights λ , the forward map is

$$\chi_i(k, \lambda) = \sum_{a=0}^3 \lambda_a X_{i,k,a}. \quad (51)$$

For inverse lookup of physical point x , define

$$D_{X,k} = [X_{i,k,1} - X_{i,k,0} \quad X_{i,k,2} - X_{i,k,0} \quad X_{i,k,3} - X_{i,k,0}]. \quad (52)$$

For each conservative candidate tetrahedron,

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = D_{X,k}^{-1}(x - X_{i,k,0}), \quad \lambda_0 = 1 - \lambda_1 - \lambda_2 - \lambda_3. \quad (53)$$

If all weights lie within the fixed tolerance, the canonical point and harmonic value are

$$\Phi_i(x) = \sum_{a=0}^3 \lambda_a Q_{k,a}, \quad (54)$$

$$r(x) = \sum_{a=0}^3 \lambda_a r_{k,a}. \quad (55)$$

The containing-tetrahedron search uses a conservative uniform AABB grid with 32 cells on the longest axis. A tetrahedron spanning more than 512 cells enters a global conservative list. Exact AABB and barycentric tests decide containment; there is no nearest-cell snapping. If several adjacent tetrahedra contain a face/edge/vertex query and reconstruct the same canonical point, the smallest ID is chosen deterministically. Disagreeing canonical reconstructions are reported as `NONINJECTIVE_OVERLAP`.

Finite points outside the tetrahedral shell are classified using the final triangulated boundaries as inside computational anatomy, outside the envelope, or numerically ambiguous. The knee truncation remains mappable but is excluded from footwear-support queries. Non-finite rows receive an independent status. The API accepts posed-SUPR, normalized-shoe, and original-shoe frames; lookup itself always occurs in posed-SUPR coordinates using the verified containment-fit transforms.

Checkpoint 11-C: identical tetrahedron IDs and barycentric weights identify corresponding points

A. Canonical volume A

B. sandal_1 volume A_i



Matching number and color = one exact canonical coordinate transported through χ_i

Figure 8: Representative exact coordinates transported from canonical volume A to the final `sandal_1` volume. Each matching label uses the same tetrahedron ID and barycentric weights; its Euclidean position changes while its canonical identity remains fixed.

9.1 Exhaustive deterministic audit

For every accepted shoe, 11-C queried five strictly interior weight patterns in every one of the 59,081 tetrahedra:

$$(0.25, 0.25, 0.25, 0.25), \quad (0.55, 0.15, 0.15, 0.15), \quad \text{and its three permutations.} \quad (56)$$

This gives 295,405 strict interior queries per shoe. The audit also checked all 14,440 volume vertices, all 127,024 unique tetrahedral face centroids, 15,584 anatomical-boundary face centroids, 860 knee-cap face centroids, 1,280 outer-boundary face centroids, and 256 samples in each external coordinate frame.

All 15 shoes report `mapping_valid`. No noninjective overlap was found. Every generated valid-domain sample was mappable and all forward/inverse, physical, canonical, harmonic- r , shared-boundary, and frame round trips passed. For `sandal_1`, the maximum strict-interior canonical round-trip error was 1.113×10^{-15} and maximum physical error was 7.027×10^{-16} in normalized computational units.

The required identities are therefore observed numerically:

$$q \xrightarrow{\chi_i} x \xrightarrow{\Phi_i} q' \approx q, \quad x \xrightarrow{\Phi_i} q \xrightarrow{\chi_i} x' \approx x. \quad (57)$$

10 Where the project is in `representation.pdf`

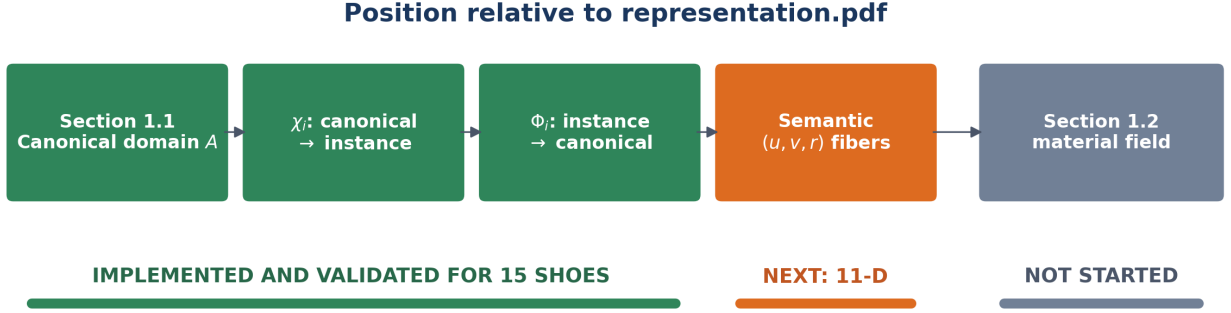


Figure 9: Current boundary between completed and pending work.

10.1 Section 1.1: what is complete

The following geometric substrate is implemented for 15 shoes:

- a canonical foot-and-lower-leg domain A ;
- one valid fitted volume $A_i = N(F_i)$ per accepted shoe;
- shared tetrahedron IDs and connectivity;
- positive, bounded-distortion piecewise-affine deformations;
- exact forward χ_i and inverse Φ_i queries;
- transported harmonic r ;
- explicit rejection/classification outside the mathematical domain.

This establishes the exact computational realization of the map proposed in Eq. (2). It also provides the shared location system required before learning any footwear prior.

10.2 Checkpoint 11-D: what remains in Section 1.1

The exact internal address in Eq. (5) is not yet the document’s semantic notation (u, v, r) . The harmonic scalar r exists, but a volume point has not yet been assigned a unique anatomical surface origin (u, v) . Checkpoint 11-D must build and validate canonical anatomical fibers

$$\gamma_{u,v}(r) \subset A, \quad \gamma_{u,v}^i(r) = \chi_i(\gamma_{u,v}(r)), \quad (58)$$

that start on anatomical skin, move monotonically outward, stay inside the valid domain, do not fold/reverse, reach the envelope as intended, and never terminate on the computational knee cap. A single flattened global UV map is not required: canonical surface face ID plus three barycentric weights can serve as the exact (u, v) chart.

10.3 Section 1.2 and beyond

Section 1.2 has not started. After 11-D, shoe triangles or material samples can be mapped through Φ_i and aggregated in shared anatomical coordinates. The proposed fields are

$$g_i(u, v), \quad \delta_i^{\text{in}}(u, v), \quad \delta_i^{\text{out}}(u, v), \quad (59)$$

with base material field

$$B_i(u, v, r) = \max\{g_i(u, v), \delta_i^{\text{in}}(u, v) - r, r - \delta_i^{\text{out}}(u, v)\}. \quad (60)$$

Learning the anatomy-conditioned clearance prior, residual material field, signed implicit footwear function, set-valued surface correspondence, and appearance/completion model belongs to Sections 1.2–1.4 and is outside the current implementation.

11 Conclusions and immediate next step

The main result is not merely that 15 meshes were processed. The result is that 15 shoes with different surface topology are now connected to one common anatomical volume while preserving their individual fitted anatomy and shoe coordinate transforms. The final volumes have fixed canonical connectivity, no folded cells, no surface intersections, bounded distortion, and reversible point lookup.

The correct next step is Checkpoint 11-D: construct one canonical, validated semantic fiber system and transport it to every instance through the already validated χ_i . Only after that should Section 1.2 convert footwear geometry into anatomical coverage and inner/outer material extents. No learning is required or appropriate before this shared semantic coordinate is complete.

A Per-shoe B3 and 11-C summary

Table 4: Final geometry and mapping metrics. Correction is normalized by fitted surface resolution h . All rows have zero final intersections and 11-C status `mapping_valid`.

Shoe	B3 result	min J	max κ	max Δ/h	11-C max. err.
AJ-12 basketball sneakers	corrected	0.0895	15.00	0.252	1.58×10^{-15}
Birkenstock Arizona sandal	exact	0.1545	9.74	0.000	9.93×10^{-16}
Canvas shoe	corrected	0.2479	5.94	0.254	9.49×10^{-16}
Crocs	corrected	0.1110	11.52	0.317	9.02×10^{-16}
Crocs (SpeedyArt)	exact	0.0820	14.04	0.000	9.17×10^{-16}
Crocs shoe	corrected	0.2783	5.43	0.258	9.16×10^{-16}
Duinn hiking sandal	exact	0.2753	6.94	0.000	9.16×10^{-16}
Leather boots	corrected	0.1027	15.00	0.257	1.86×10^{-15}
Nike Air Jordan	corrected	0.0940	14.95	0.257	1.16×10^{-15}
PB129 low shoe	exact	0.0735	15.00	0.000	1.98×10^{-15}
Priest sports shoe	corrected	0.3087	4.45	0.234	9.17×10^{-16}

Shoe	B3 result	$\min J$	$\max \kappa$	$\max \Delta/h$	11-C max. err.
Sandal 1	corrected	0.0957	9.36	0.258	1.11×10^{-15}
Sandals 0001	corrected	0.3053	6.40	0.255	1.11×10^{-15}
Sneakers Seen	corrected	0.1213	14.54	0.258	9.17×10^{-16}
WWII German jack boots	corrected	0.0878	12.69	0.229	1.30×10^{-15}

B Principal artifact locations

Previously reported support fits

/home/ab5298/Outputs/FootShellGaussian/golden_set_evaluation/support_fit

Cavity diagnoses

/home/ab5298/Outputs/FootShellGaussian/golden_set_evaluation/cavity_analysis

Containment-fitted anatomy

/home/ab5298/Outputs/FootShellGaussian/golden_set_evaluation/containment_fit

Dense and extended anatomy

/home/ab5298/Outputs/FootShellGaussian/golden_set_evaluation/anatomical_surface

/home/ab5298/Outputs/FootShellGaussian/golden_set_evaluation/extended_anatomical_surface

Canonical volume and 11-A targets

/home/ab5298/Outputs/FootShellGaussian/golden_set_evaluation/anatomical_volume

Accepted 15-shoe B3 batch

/home/ab5298/Outputs/FootShellGaussian/golden_set_evaluation/instance_anatomical_volume/batch_threads1_20260913_181358

Accepted 15-shoe 11-C audit

/home/ab5298/Outputs/FootShellGaussian/golden_set_evaluation/instance_volume_mapping/batch_threads1_20260913_181358

Representation design document

/storage/Abhinay/Shell_Gaussian/notes/representation.pdf

C Reproducibility note

The numerical statements in this report were read from the saved JSON/NPZ artifacts and the implemented fixed configurations. The figures were rendered directly from the saved PLY/NPZ meshes and fields using `notes/shared_anatomical_volume_report_assets/generate_figures.py`. The B3 batch manifest records the exact source digests and one-thread numerical-library policy used for the accepted run. No B3 or mapping artifact was modified while preparing this report.